spinning bucket in an otherwise empty universe. In an empty unchanging universe—no stars, no planets, no anything at all—there is no gravity.[21] And without gravity, spacetime is not warped—it takes the simple, uncurved shape shown in Figure 3.9b—and that means we are back in the simpler setting of special relativity. (Remember, Einstein ignored gravity while developing special relativity. General relativity made up for this deficiency by incorporating gravity, but when the universe is empty and unchanging there is no gravity, and so general relativity reduces to special relativity.) If we now introduce the bucket into this empty universe, it has such a tiny mass that its presence hardly affects the shape of space at all. And so the discussion we had earlier for the bucket in special relativity applies equally well to general relativity. In contradiction to what Mach would have predicted, general relativity comes to the same answer as special relativity, and proclaims that even in an otherwise empty universe, you *will* feel pressed against the inner wall of the spinning bucket; in an otherwise empty universe, your arms *will* feel pulled outward if you spin around; in an otherwise empty universe, the rope tied between two twirling rocks *will* become taut. The conclusion we draw is that even in general relativity, empty spacetime provides a benchmark for accelerated motion.

Hence, although general relativity incorporates some elements of Mach's thinking, it does not subscribe to the completely relative conception of motion Mach advocated.[22] Mach's principle is an example of a provocative idea that provided inspiration for a revolutionary discovery even though that discovery ultimately failed to fully embrace the idea that inspired it.

Spacetime in the Third Millennium

The spinning bucket has had a long run. From Newton's absolute space and absolute time, to Leibniz's and then Mach's relational conceptions, to Einstein's realization in special relativity that space and time are relative and yet in their union fill out absolute spacetime, to his subsequent discovery in general relativity that spacetime is a dynamic player in the unfolding cosmos, the bucket has always been there. Twirling in the back of the mind, it has provided a simple and quiet test for whether the invisible, the abstract, the untouchable stuff of space—